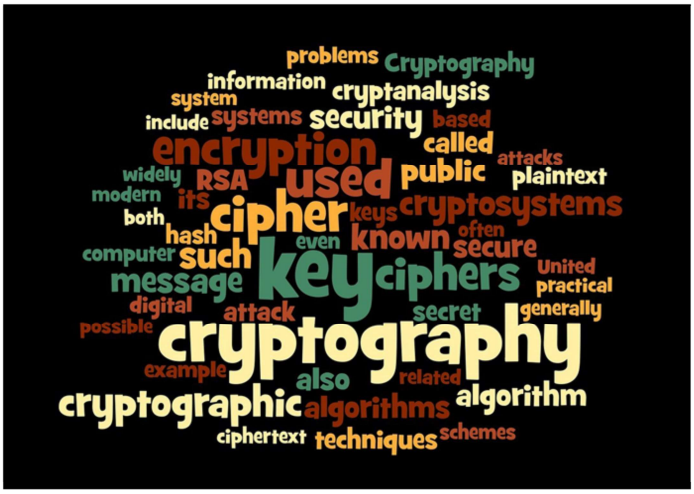
Mitigations

- Isolated on private network
- Products selected for security features and updatability
- Product security/pentesting
- Disable unneeded functions
- Cohesive Digital Forensics and Incident Response controls/procedures

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Cryptography role in Security



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Reference https://en.wikipedia.org/wiki/Cryptography

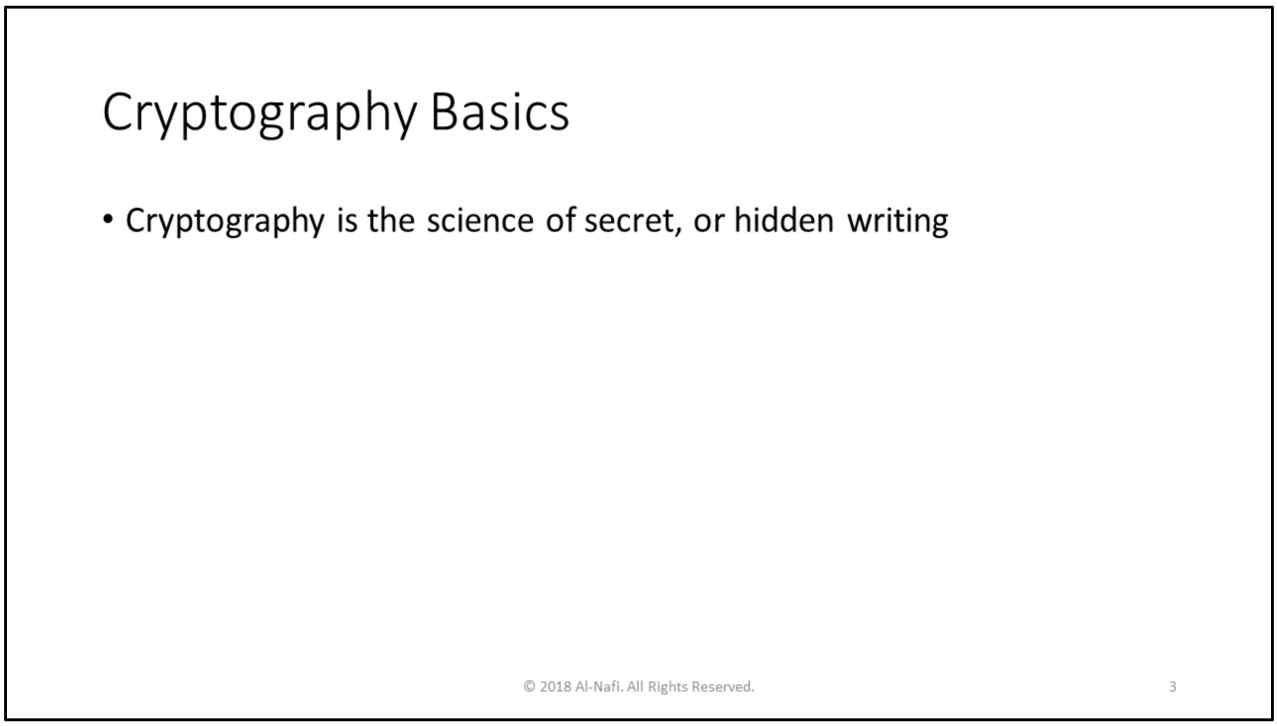
Cryptography or **cryptology** (from Ancient Greek: κρυπτός, romanized: *kryptós* "hidden, secret"; and γράφειν *graphein*, "to write", or -λογία *-logia*, "study", respectively[1]) is the practice and study of techniques for secure communication in the presence of third parties called adversaries.[2] More generally, cryptography is about constructing and analyzing protocols that prevent third parties or the public from reading private messages;[3] various aspects in information security such as data confidentiality, data integrity, authentication, and non-repudiation[4] are central to modern cryptography. Modern cryptography exists at the intersection of the disciplines of mathematics, computer science, electrical engineering, communication science, and physics. Applications of cryptography include electronic commerce, chip-based payment cards, digital currencies, computer passwords, and military communications.

Cryptography prior to the modern age was effectively synonymous with *encryption*, the conversion of information from a readable state to apparent nonsense. The

originator of an encrypted message shares the decoding technique only with intended recipients to preclude access from adversaries. The cryptography literature often uses the names Alice ("A") for the sender, Bob ("B") for the intended recipient, and Eve ("eavesdropper") for the adversary.[5] Since the development of rotor cipher machines in World War I and the advent of computers in World War II, the methods used to carry out cryptology have become increasingly complex and its application more widespread.

Modern cryptography is heavily based on mathematical theory and computer science practice; cryptographic algorithms are designed around computational hardness assumptions, making such algorithms hard to break in practice by any adversary. It is theoretically possible to break such a system, but it is infeasible to do so by any known practical means. These schemes are therefore termed computationally secure; theoretical advances, e.g., improvements in integer factorization algorithms, and faster computing technology require these solutions to be continually adapted. There exist information-theoretically secure schemes that provably cannot be broken even with unlimited computing power—an example is the one-time pad—but these schemes are more difficult to use in practice than the best theoretically breakable but computationally secure mechanisms.

The growth of cryptographic technology has raised a number of legal issues in the information age. Cryptography's potential for use as a tool for espionage and sedition has led many governments to classify it as a weapon and to limit or even prohibit its use and export.[6] In some jurisdictions where the use of cryptography is legal, laws permit investigators to compel the disclosure of encryption keys for documents relevant to an investigation.[7][8] Cryptography also plays a major role in digital rights management and copyright infringement of digital media.[9]



Reference https://en.wikipedia.org/wiki/Cryptography

Terminology

The first use of the term *cryptograph* (as opposed to *cryptogram*) dates back to the 19th century—originating from *The Gold-Bug*, a novel by Edgar Allan Poe.[10][11]

Until modern times, cryptography referred almost exclusively to *encryption*, which is the process of converting ordinary information (called plaintext) into unintelligible form (called ciphertext).[12] Decryption is the reverse, in other words, moving from the unintelligible ciphertext back to plaintext.

A *cipher* (or *cypher*) is a pair of algorithms that create the encryption and the reversing decryption. The detailed operation of a cipher is controlled both by the algorithm and in each instance by a "key". The key is a secret (ideally known only to the communicants), usually a short string of characters, which is needed to decrypt the ciphertext. Formally, a "cryptosystem" is the ordered list of elements of finite possible plaintexts, finite possible cyphertexts, finite possible keys, and the

encryption and decryption algorithms which correspond to each key. Keys are important both formally and in actual practice, as ciphers without variable keys can be trivially broken with only the knowledge of the cipher used and are therefore useless (or even counter-productive) for most purposes.

Historically, ciphers were often used directly for encryption or decryption without additional procedures such as authentication or integrity checks. There are two kinds of cryptosystems: symmetric and asymmetric. In symmetric systems the same key (the secret key) is used to encrypt and decrypt a message. Data manipulation in symmetric systems is faster than asymmetric systems as they generally use shorter key lengths. Asymmetric systems use a public key to encrypt a message and a private key to decrypt it. Use of asymmetric systems enhances the security of communication.[13] Examples of asymmetric systems include RSA (Rivest–Shamir–Adleman), and ECC (Elliptic Curve Cryptography). Symmetric models include the commonly used AES (Advanced Encryption Standard) which replaced the older DES (Data Encryption Standard).[14]

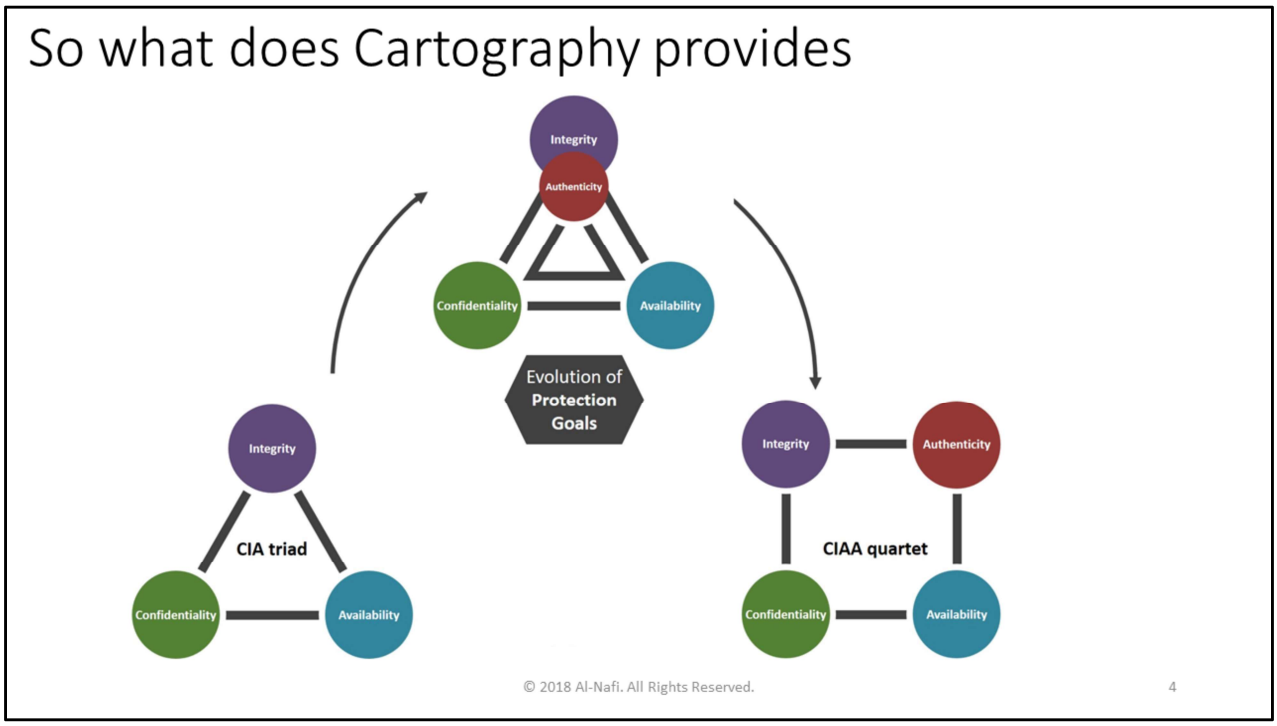
In colloquial use, the term "code" is often used to mean any method of encryption or concealment of meaning. However, in cryptography, *code* has a more specific meaning. It means the replacement of a unit of plaintext (i.e., a meaningful word or phrase) with a code word (for example, "wallaby" replaces "attack at dawn").

Cryptanalysis is the term used for the study of methods for obtaining the meaning of encrypted information without access to the key normally required to do so; i.e., it is the study of how to crack encryption algorithms or their implementations.

Some use the terms *cryptography* and *cryptology* interchangeably in English, while others (including US military practice generally) use *cryptography* to refer specifically to the use and practice of cryptographic techniques and *cryptology* to refer to the combined study of cryptography and cryptanalysis.[15][16] English is more flexible than several other languages in which *cryptology* (done by cryptologists) is always used in the second sense above. RFC 2828 advises that steganography is sometimes included in cryptology.[17]
The study of characteristics of languages that have some application in cryptography or cryptology (e.g. frequency data, letter combinations, universal patterns, etc.) is called cryptolinguistics.

History of cryptography and cryptanalysis
Main article: History of cryptography
Before the modern era, cryptography focused on message confidentiality (i.e., encryption)—conversion of messages from a comprehensible form into an

incomprehensible one and back again at the other end, rendering it unreadable by interceptors or eavesdroppers without secret knowledge (namely the key needed for decryption of that message). Encryption attempted to ensure secrecy in communications, such as those of spies, military leaders, and diplomats. In recent decades, the field has expanded beyond confidentiality concerns to include techniques for message integrity checking, sender/receiver identity authentication, digital signatures, interactive proofs and secure computation, among others.



Cryptography today can be said to provide some important security services. The five key services that cryptography can provide are the following:

Confidentiality: Cryptography provides confidentiality through altering or hiding a message so that ideally it cannot be understood by anyone except the intended recipient. Confidentiality is a service that ensures keeping information secret from those who are not authorized to have it. Secrecy is a term sometimes used to mean confidentiality.

Integrity: Cryptographic tools can provide integrity services that allow a recipient to verify that a message has not been altered. Cryptography tools cannot prevent a message from being altered, but they can be effective to detect either intentional or accidental modification of the message.

Cryptographic functions use several methods to ensure that a message has not been changed or altered. These may include hash functions, digital signatures, and simpler message integrity controls such a message authentication codes (MACs),

Cyclic Redundancy Checks (CRC), or even checksums. The concept behind this is that the recipient is able to detect any change that has been made to a message, whether accidentally or intentionally.

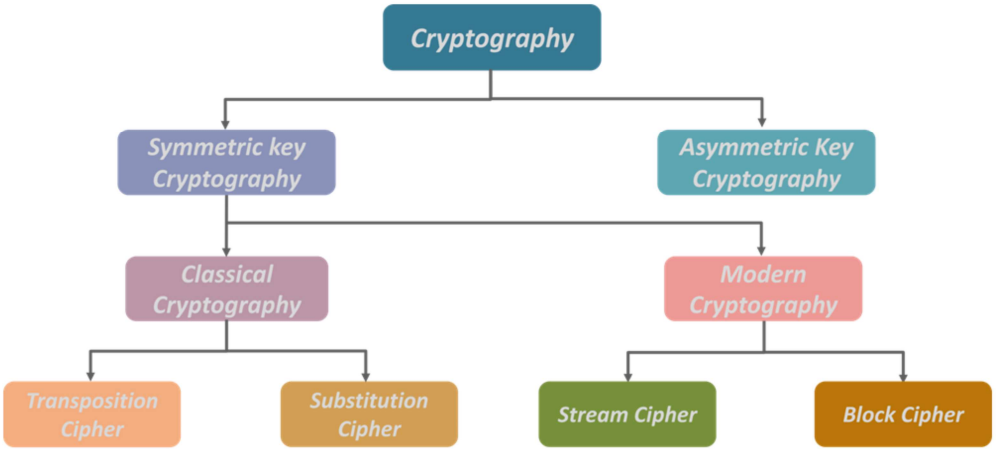
Authenticity: Sometimes referred to as “proof of origin,” this is a service that allows entities wanting to communicate with each other to positively identify each other. Information delivered over a channel should be authenticated as to the origin of that transmission. Authenticity can allow a recipient to know positively that a transmission of information actually came from the entity that we expect it from.

Non-repudiation: This is a service that prevents an entity from denying having participated in a previous action. Typically, nonrepudiation can only be achieved properly through the use of digital signatures. The word *repudiation* means the ability to deny. So, non-repudiation means the inability to deny. There are two flavors of non-repudiation:

- Non-repudiation of origin means that the sender cannot deny they sent a particular message.
- Non-repudiation of delivery means that the receiver cannot say that they received a different message than the one they actually did receive.

Access Control: This is an added benefit of Cryptography. Through the use of cryptographic tools, many forms of access control are supported—from log-ins via passwords and passphrases to the prevention of access to confidential files or messages. In all cases, access would only be possible for those individuals who had access to the correct cryptographic keys.

Types of Cryptography



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We will cover them in detail later in the chapter inshAllah.

What is plain text v.s cipher text

- Lets do this online https://cryptii.com/pipes/caesar-cipher

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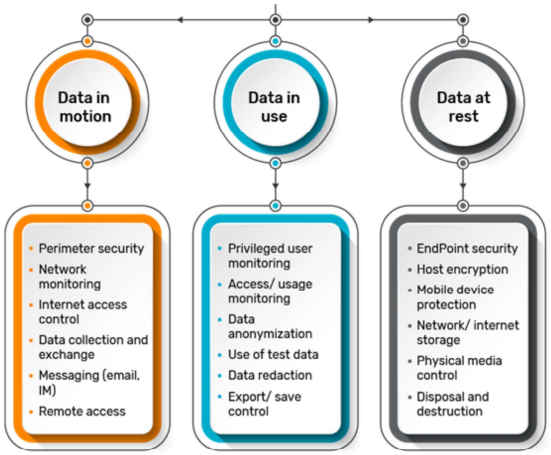
As we have seen, cryptography is about writing secrets. The first secret messages were exchanged as long as thousands of years ago. Cryptography involves scrambling some kind of useful information in its original form, called plaintext, into a garbled or secret form, called ciphertext. The usual intent is to allow two or more parties to communicate the information while preventing other parties from being privy to it.

Data Protection



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Data in motion, in use and at rest



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Data at Rest

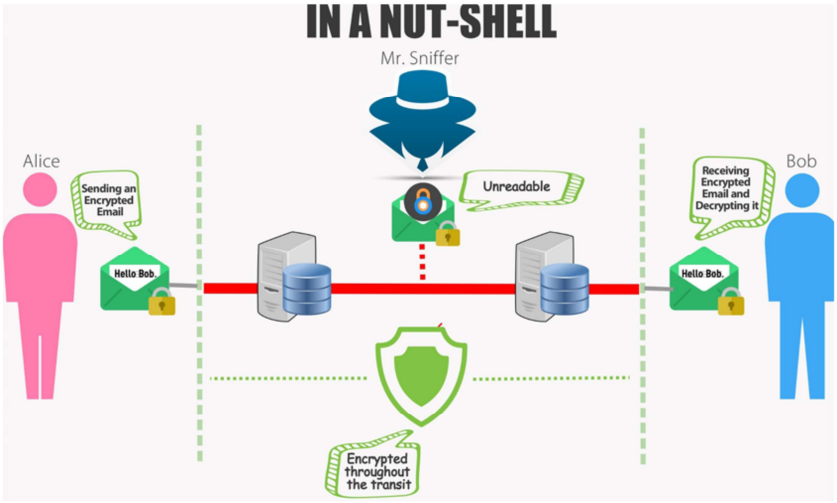
The protection of stored data is often a key requirement for an organization's sensitive information. Backups, off-site storage, password files, sensitive databases, valuable files, and other types of sensitive information need to be protected from disclosure or undetected alteration. This can usually be done through the use of cryptographic algorithms that limit access to the data to those that hold the proper encryption (and decryption) keys. Protecting these valuable examples of assets of the organization can be done usually through cryptography, but it is usually referred to as protecting data at rest. Data at rest means the data is resting, stored on some storage media without it moving at any point.

Data in Transit or in motion aka data is on the move

Data in transit, sometimes referred to as data in motion, is data that is moving, usually across networks. Whether the message is sent manually, over a voice network, or via the internet, modern cryptography can provide secure and confidential methods to transmit data and allows the verification of the integrity of

the
message so that any changes to the message itself can be detected.

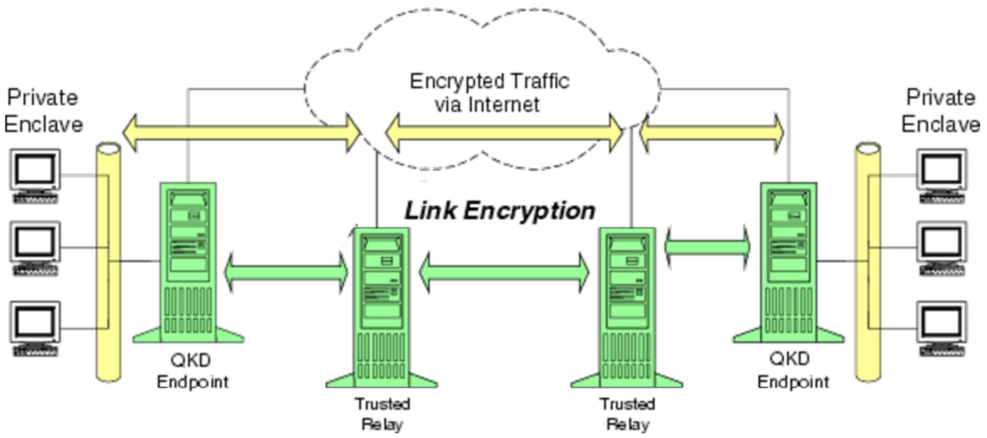
End to End Encryption



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End-to-end encryption is generally performed by the end user within an organization. The data is encrypted at the start of the communications channel or before and remains encrypted until it is
decrypted at the remote end. Although data remain encrypted when passed through a network, routing information remains visible.

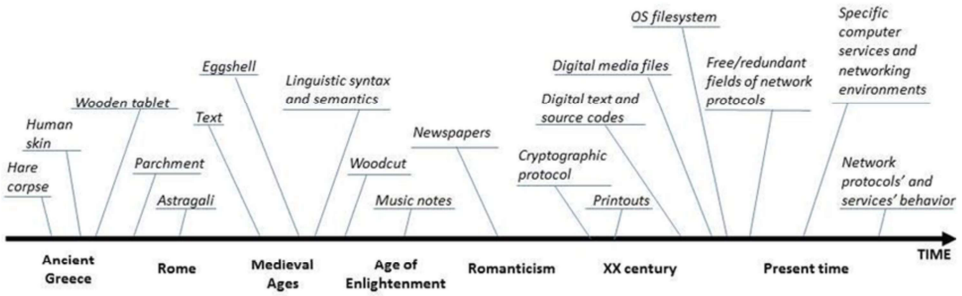
Link Encryption



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Data that is moving across a network can be protected using cryptography. There are two methods for protecting data in transit across a network, link or end-to-end encryption. In general, link encryption is performed by service providers, such as a data communications provider on networks. Link encryption encrypts all of the data along a communications path (e.g., a satellite link, telephone circuit, or T-1 line). Because link encryption also encrypts routing data, communications nodes need to decrypt the data to
continue routing. The data packet is decrypted and re-encrypted at each point in the communications channel. It is theoretically possible that an attacker compromising a node in the network may see the message in the clear. Because link encryption also encrypts the routing information, it provides traffic confidentiality (not data confidentiality) better than end-to end encryption. In other words, it can be used to hide the routing information. Traffic confidentiality hides the addressing information from an observer, preventing an inference attack based on the existence of traffic between two parties.

Cryptographic Evolution



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Reference https://en.wikipedia.org/wiki/History_of_cryptography

Please read this in detail

Cryptography, the use of codes and ciphers to protect secrets, began thousands of years ago. Until recent decades, it has been the story of what might be called classic cryptography — that is, of methods of encryption that use pen and paper, or perhaps simple mechanical aids. In the early 20th century, the invention of complex mechanical and electromechanical machines, such as the Enigma rotor machine, provided more sophisticated and efficient means of encryption; and the subsequent introduction of electronics and computing has allowed elaborate schemes of still greater complexity, most of which are entirely unsuited to pen and paper.

The development of cryptography has been paralleled by the development of cryptanalysis — the "breaking" of codes and ciphers. The discovery and application, early on, of frequency analysis to the reading of encrypted communications has, on occasion, altered the course of history. Thus the Zimmermann Telegram triggered the United States' entry into World War I;

and Allied reading of Nazi Germany's ciphers shortened World War II, in some evaluations by as much as two years.
Until the 1970s, secure cryptography was largely the preserve of governments. Two events have since brought it squarely into the public domain: the creation of a public encryption standard (DES), and the invention of public-key cryptography.

Antiquity

A Scytale, an early device for encryption.
The earliest known use of cryptography is found in non-standard hieroglyphs carved into the wall of a tomb from the Old Kingdom of Egypt circa 1900 BC.[1] These are not thought to be serious attempts at secret communications, however, but rather to have been attempts at mystery, intrigue, or even amusement for literate onlookers.[1][*failed verification*]
Some clay tablets from Mesopotamia somewhat later are clearly meant to protect information—one dated near 1500 BC was found to encrypt a craftsman's recipe for pottery glaze, presumably commercially valuable.[2][3] Furthermore, Hebrew scholars made use of simple monoalphabetic substitution ciphers (such as the Atbash cipher) beginning perhaps around 500 to 600 BC.[4][5]
In India around 400 BC to 200 AD, Mlecchita vikalpa or *the art of understanding writing in cypher, and the writing of words in a peculiar way* was documented in the Kama Sutra for the purpose of communication between lovers. This was also likely a simple substitution cipher.[6][7] Parts of the Egyptian demotic Greek Magical Papyri were written in a cypher script.[8]
The ancient Greeks are said to have known of ciphers.[9] The scytale transposition cipher was used by the Spartan military,[5] but it is not definitively known whether the scytale was for encryption, authentication, or avoiding bad omens in speech.[10][11] Herodotus tells us of secret messages physically concealed beneath wax on wooden tablets or as a tattoo on a slave's head concealed by regrown hair, although these are not properly examples of cryptography *per se* as the message, once known, is directly readable; this is known as steganography. Another Greek method was developed by Polybius (now called the "Polybius Square").[5] The Romans knew something of cryptography (e.g., the Caesar cipher and its variations).[12]

Medieval cryptography

The first page of al-Kindi's manuscript *On Deciphering Cryptographic Messages*, containing the first descriptions of cryptanalysis and frequency analysis.
See also: Classical cipher and Voynich Manuscript
David Kahn notes in *The Codebreakers* that modern cryptology originated among the Arabs, the first people to systematically document cryptanalytic methods.[13] Al-

Khalil (717–786) wrote the *Book of Cryptographic Messages*, which contains the first use of permutations and combinations to list all possible Arabic words with and without vowels.[14]

The invention of the frequency analysis technique for breaking monoalphabetic substitution ciphers, by Al-Kindi, an Arab mathematician,[15][16] sometime around AD 800, proved to be the single most significant cryptanalytic advance until World War II. Al-Kindi wrote a book on cryptography entitled *Risalah fi Istikhraj al-Mu'amma* (*Manuscript for the Deciphering Cryptographic Messages*), in which he described the first cryptanalytic techniques, including some for polyalphabetic ciphers, cipher classification, Arabic phonetics and syntax, and most importantly, gave the first descriptions on frequency analysis.[17] He also covered methods of encipherments, cryptanalysis of certain encipherments, and statistical analysis of letters and letter combinations in Arabic.[18][19] An important contribution of Ibn Adlan (1187–1268) was on sample size for use of frequency analysis.[14]

In early medieval England between the years 800-1100, substitution ciphers were frequently used by scribes as a playful and clever way encipher notes, solutions to riddles, and colophons. The ciphers tend to be fairly straightforward, but sometimes they deviate from an ordinary pattern, adding to their complexity and, possibly, to their sophistication as well.[20] This period saw vital and significant cryptographic experimentation in the West.

Ahmad al-Qalqashandi (AD 1355–1418) wrote the *Subh al-a 'sha*, a 14-volume encyclopedia which included a section on cryptology. This information was attributed to Ibn al-Durayhim who lived from AD 1312 to 1361, but whose writings on cryptography have been lost. The list of ciphers in this work included both substitution and transposition, and for the first time, a cipher with multiple substitutions for each plaintext letter. Also traced to Ibn al-Durayhim is an exposition on and worked example of cryptanalysis, including the use of tables of letter frequencies and sets of letters which cannot occur together in one word.

The earliest example of the homophonic substitution cipher is the one used by Duke of Mantua in the early 1400s.[21] Homophonic cipher replaces each letter with multiple symbols depending on the letter frequency. The cipher is ahead of the time because it combines monoalphabetic and polyalphabetic features.

Essentially all ciphers remained vulnerable to the cryptanalytic technique of frequency analysis until the development of the polyalphabetic cipher, and many remained so thereafter. The polyalphabetic cipher was most clearly explained

by Leon Battista Alberti around the year AD 1467, for which he was called the "father of Western cryptology".[1] Johannes Trithemius, in his work Poligraphia, invented the tabula recta, a critical component of the Vigenère cipher. Trithemius also wrote the Steganographia. The French cryptographer Blaise de Vigenère devised a practical polyalphabetic system which bears his name, the Vigenère cipher.[1]

In Europe, cryptography became (secretly) more important as a consequence of political competition and religious revolution. For instance, in Europe during and after the Renaissance, citizens of the various Italian states—the Papal States and the Roman Catholic Church included—were responsible for rapid proliferation of cryptographic techniques, few of which reflect understanding (or even knowledge) of Alberti's polyalphabetic advance. 'Advanced ciphers', even after Alberti, weren't as advanced as their inventors / developers / users claimed (and probably even themselves believed). They were regularly broken. This over-optimism may be inherent in cryptography, for it was then - and remains today - fundamentally difficult to accurately know how vulnerable one's system actually is. In the absence of knowledge, guesses and hopes, predictably, are common.

Cryptography, cryptanalysis, and secret-agent/courier betrayal featured in the Babington plot during the reign of Queen Elizabeth I which led to the execution of Mary, Queen of Scots. Robert Hooke suggested in the chapter *Of Dr. Dee's Book of Spirits*, that John Dee made use of Trithemian steganography, to conceal his communication with Queen Elizabeth I.[22]

The chief cryptographer of King Louis XIV of France was Antoine Rossignol and he and his family created what is known as the Great Cipher because it remained unsolved from its initial use until 1890, when French military cryptanalyst, Étienne Bazeries solved it.[23] An encrypted message from the time of the Man in the Iron Mask (decrypted just prior to 1900 by Étienne Bazeries) has shed some, regrettably non-definitive, light on the identity of that real, if legendary and unfortunate, prisoner.
Outside of Europe, after the Mongols brought about the end of the Islamic Golden Age, cryptography remained comparatively undeveloped. Cryptography in Japan seems not to have been used until about 1510, and advanced techniques were not known until after the opening of the country to the West beginning in the 1860s.

Cryptography from 1800 to World War II

Main article: World War I cryptography
Although cryptography has a long and complex history, it wasn't until the 19th

century that it developed anything more than ad hoc approaches to either encryption or cryptanalysis (the science of finding weaknesses in crypto systems). Examples of the latter include Charles Babbage's Crimean War era work on mathematical cryptanalysis of polyalphabetic ciphers, redeveloped and published somewhat later by the Prussian Friedrich Kasiski. Understanding of cryptography at this time typically consisted of hard-won rules of thumb; see, for example, Auguste Kerckhoffs' cryptographic writings in the latter 19th century. Edgar Allan Poe used systematic methods to solve ciphers in the 1840s. In particular he placed a notice of his abilities in the Philadelphia paper *Alexander's Weekly (Express) Messenger*, inviting submissions of ciphers, of which he proceeded to solve almost all. His success created a public stir for some months.[24] He later wrote an essay on methods of cryptography which proved useful as an introduction for novice British cryptanalysts attempting to break German codes and ciphers during World War I, and a famous story, *The Gold-Bug*, in which cryptanalysis was a prominent element.
Cryptography, and its misuse, were involved in the execution of Mata Hari and in Dreyfus' conviction and imprisonment, both in the early 20th century. Cryptographers were also involved in exposing the machinations which had led to the Dreyfus affair; Mata Hari, in contrast, was shot.
In World War I the Admiralty's Room 40 broke German naval codes and played an important role in several naval engagements during the war, notably in detecting major German sorties into the North Sea that led to the battles of Dogger Bank and Jutland as the British fleet was sent out to intercept them. However its most important contribution was probably in decrypting the Zimmermann Telegram, a cable from the German Foreign Office sent via Washington to its ambassador Heinrich von Eckardt in Mexico which played a major part in bringing the United States into the war.
In 1917, Gilbert Vernam proposed a teleprinter cipher in which a previously prepared key, kept on paper tape, is combined character by character with the plaintext message to produce the cyphertext. This led to the development of electromechanical devices as cipher machines, and to the only unbreakable cipher, the one time pad.
During the 1920s, Polish naval-officers assisted the Japanese military with code and cipher development.
Mathematical methods proliferated in the period prior to World War II (notably in William F. Friedman's application of statistical techniques to cryptanalysis and cipher development and in Marian Rejewski's initial break into the German Army's version of the Enigma system in 1932).

World War II cryptography[edit]

See also: World War II cryptography, Cryptanalysis, and List of cryptographers
The Enigma machine was widely used by Nazi Germany; its cryptanalysis by the Allies

provided vital Ultra intelligence.
By World War II, mechanical and electromechanical cipher machines were in wide use, although—where such machines were impractical—code books and manual systems continued in use. Great advances were made in both cipher design and cryptanalysis, all in secrecy. Information about this period has begun to be declassified as the official British 50-year secrecy period has come to an end, as US archives have slowly opened, and as assorted memoirs and articles have appeared.

Germany[edit]

The Germans made heavy use, in several variants, of an electromechanical rotor machine known as Enigma.[25] Mathematician Marian Rejewski, at Poland's Cipher Bureau, in December 1932 deduced the detailed structure of the German Army Enigma, using mathematics and limited documentation supplied by Captain Gustave Bertrand of French military intelligence. This was the greatest breakthrough in cryptanalysis in a thousand years and more, according to historian David Kahn.[*citation needed*] Rejewski and his mathematical Cipher Bureau colleagues, Jerzy Różycki and Henryk Zygalski, continued reading Enigma and keeping pace with the evolution of the German Army machine's components and encipherment procedures. As the Poles' resources became strained by the changes being introduced by the Germans, and as war loomed, the Cipher Bureau, on the Polish General Staff's instructions, on 25 July 1939, at Warsaw, initiated French and British intelligence representatives into the secrets of Enigma decryption.
Soon after the Invasion of Poland by Germany on 1 September 1939, key Cipher Bureau personnel were evacuated southeastward; on 17 September, as the Soviet Union attacked Poland from the East, they crossed into Romania. From there they reached Paris, France; at PC Bruno, near Paris, they continued breaking Enigma, collaborating with British cryptologists at Bletchley Park as the British got up to speed on breaking Enigma. In due course, the British cryptographers – whose ranks included many chess masters and mathematics dons such as Gordon Welchman, Max Newman, and Alan Turing (the conceptual founder of modern computing) – substantially advanced the scale and technology of Enigma decryption.
German code breaking in World War II also had some success, most importantly by breaking the Naval Cipher No. 3. This enabled them to track and sink Atlantic convoys. It was only Ultra intelligence that finally persuaded the admiralty to change their codes in June 1943. This is surprising given the success of the British Room 40 code breakers in the previous world war.
At the end of the War, on 19 April 1945, Britain's top military officers were told that they could never reveal that the German Enigma cipher had been broken because it would give the defeated enemy the chance to say they "were not well and fairly beaten".[26]
The German military also deployed several teleprinter stream ciphers. Bletchley Park called them the Fish ciphers, and Max Newman and colleagues designed and

deployed the Heath Robinson, and then the world's first programmable digital electronic computer, the Colossus, to help with their cryptanalysis. The German Foreign Office began to use the one-time pad in 1919; some of this traffic was read in World War II partly as the result of recovery of some key material in South America that was discarded without sufficient care by a German courier.
The *Schlüsselgerät 41* was developed late in the war as a more secure replacement for Enigma, but only saw limited use.

Japan[edit]

A US Army group, the SIS, managed to break the highest security Japanese diplomatic cipher system (an electromechanical stepping switch machine called Purple by the Americans) in 1940, before World War II began. The locally developed Purple machine replaced the earlier "Red" machine used by the Japanese Foreign Ministry, and a related machine, the M-1, used by Naval attachés which was broken by the U.S. Navy's Agnes Driscoll. All the Japanese machine ciphers were broken, to one degree or another, by the Allies.
The Japanese Navy and Army largely used code book systems, later with a separate numerical additive. US Navy cryptographers (with cooperation from British and Dutch cryptographers after 1940) broke into several Japanese Navy crypto systems. The break into one of them, JN-25, famously led to the US victory in the Battle of Midway; and to the publication of that fact in the Chicago Tribune shortly after the battle, though the Japanese seem not to have noticed for they kept using the JN-25 system.

Allies[edit]

The Americans referred to the intelligence resulting from cryptanalysis, perhaps especially that from the Purple machine, as 'Magic'. The British eventually settled on 'Ultra' for intelligence resulting from cryptanalysis, particularly that from message traffic protected by the various Enigmas. An earlier British term for Ultra had been 'Boniface' in an attempt to suggest, if betrayed, that it might have an individual agent as a source.
SIGABA is described in U.S. Patent 6,175,625, filed in 1944 but not issued until 2001. Allied cipher machines used in World War II included the British TypeX and the American SIGABA; both were electromechanical rotor designs similar in spirit to the Enigma, albeit with major improvements. Neither is known to have been broken by anyone during the War. The Poles used the Lacida machine, but its security was found to be less than intended (by Polish Army cryptographers in the UK), and its use was discontinued. US troops in the field used the M-209 and the still less secure M-94 family machines. British SOE agents initially used 'poem ciphers' (memorized poems were the encryption/decryption keys), but later in the War, they began to switch to one-time pads.
The VIC cipher (used at least until 1957 in connection with Rudolf Abel's NY spy ring) was a very complex hand cipher, and is claimed to be the most complicated known to have been used by the Soviets, according to David Kahn in *Kahn on Codes*. For the

decrypting of Soviet ciphers (particularly when *one-time pads* were reused), see Venona project.

Role of women[edit]

The UK and US employed large numbers of women in their code-breaking operation, with close to 7,000 reporting to Bletchley Park[27] and 11,000 to the separate US Army and Navy operations, around Washington, DC.[28] By tradition in Japan and Nazi doctrine in Germany, women were excluded from war work, at least until late in the war. Even after encryption systems were broken, large amounts of work was needed to respond to changes made, recover daily key stettings for multiple networks, and intercept, process, translate, prioritize and analyze the huge volume of enemy messages generated in a global conflict. A few women, including Elizabeth Friedman and Agnes Meyer Driscoll, had been major contributors to US code breaking in the 1930s and the Navy and Army began actively recuiting top graduates of women's colleges shortly before the attack on Pearl Harbor. Liza Mundy argues that this disparity in utilizing the talents of women between the Allies and Axis made a strategic difference in the war.[28]:p.29

Modern cryptography[edit]

Encryption in modern times is achieved by using algorithms that have a key to encrypt and decrypt information. These keys convert the messages and data into "digital gibberish" through encryption and then return them to the original form through decryption. In general, the longer the key is, the more difficult it is to crack the code. This holds true because deciphering an encrypted message by brute force would require the attacker to try every possible key. To put this in context, each binary unit of information, or bit, has a value of 0 or 1. An 8-bit key would then have 256 or 2^8 possible keys. A 56-bit key would have 2^56, or 72 quadrillion, possible keys to try and decipher the message. With modern technology, cyphers using keys with these lengths are becoming easier to decipher. DES, an early US Government approved cypher, has an effective key length of 56 bits, and test messages using that cypher have been broken by brute force key search. However, as technology advances, so does the quality of encryption. Since World War II, one of the most notable advances in the study of cryptography is the introduction of the asymmetric key cyphers (sometimes termed public-key cyphers). These are algorithms which use two mathematically related keys for encryption of the same message. Some of these algorithms permit publication of one of the keys, due to it being extremely difficult to determine one key simply from knowledge of the other.[29]

Beginning around 1990, the use of the Internet for commercial purposes and the introduction of commercial transactions over the Internet called for a widespread standard for encryption. Before the introduction of the Advanced Encryption Standard (AES), information sent over the Internet, such as financial data, was encrypted if at all, most commonly using the Data Encryption Standard (DES). This

had been approved by NBS (a US Government agency) for its security, after public call for, and a competition among, candidates for such a cypher algorithm. DES was approved for a short period, but saw extended use due to complex wrangles over the use by the public of high quality encryption. DES was finally replaced by the AES after another public competition organized by the NBS successor agency, NIST. Around the late 1990s to early 2000s, the use of public-key algorithms became a more common approach for encryption, and soon a hybrid of the two schemes became the most accepted way for e-commerce operations to proceed. Additionally, the creation of a new protocol known as the Secure Socket Layer, or SSL, led the way for online transactions to take place. Transactions ranging from purchasing goods to online bill pay and banking used SSL. Furthermore, as wireless Internet connections became more common among households, the need for encryption grew, as a level of security was needed in these everyday situations.[30]

Claude Shannon[edit]

Claude E. Shannon is considered by many[*weasel words*] to be the father of mathematical cryptography. Shannon worked for several years at Bell Labs, and during his time there, he produced an article entitled "A mathematical theory of cryptography". This article was written in 1945 and eventually was published in the Bell System Technical Journal in 1949.[31] It is commonly accepted that this paper was the starting point for development of modern cryptography. Shannon was inspired during the war to address "[t]he problems of cryptography [because] secrecy systems furnish an interesting application of communication theory". Shannon identified the two main goals of cryptography: secrecy and authenticity. His focus was on exploring secrecy and thirty-five years later, G.J. Simmons would address the issue of authenticity. Shannon wrote a further article entitled "A mathematical theory of communication" which highlights one of the most significant aspects of his work: cryptography's transition from art to science.[32]

In his works, Shannon described the two basic types of systems for secrecy. The first are those designed with the intent to protect against hackers and attackers who have infinite resources with which to decode a message (theoretical secrecy, now unconditional security), and the second are those designed to protect against hackers and attacks with finite resources with which to decode a message (practical secrecy, now computational security). Most of Shannon's work focused around theoretical secrecy; here, Shannon introduced a definition for the "unbreakability" of a cipher. If a cipher was determined "unbreakable", it was considered to have "perfect secrecy". In proving "perfect secrecy", Shannon determined that this could only be obtained with a secret key whose length given in binary digits was greater than or equal to the number of bits contained in the information being encrypted. Furthermore, Shannon developed the "unicity distance", defined as the "amount of plaintext that… determines the secret key."[32]

Shannon's work influenced further cryptography research in the 1970s, as the public-

key cryptography developers, M. E. Hellman and W. Diffie cited Shannon's research as a major influence. His work also impacted modern designs of secret-key ciphers. At the end of Shannon's work with cryptography, progress slowed until Hellman and Diffie introduced their paper involving "public-key cryptography".[32]

An encryption standard[edit]

The mid-1970s saw two major public (i.e., non-secret) advances. First was the publication of the draft Data Encryption Standard in the U.S. *Federal Register* on 17 March 1975. The proposed DES cipher was submitted by a research group at IBM, at the invitation of the National Bureau of Standards (now NIST), in an effort to develop secure electronic communication facilities for businesses such as banks and other large financial organizations. After advice and modification by the NSA, acting behind the scenes, it was adopted and published as a Federal Information Processing Standard Publication in 1977 (currently at FIPS 46-3). DES was the first publicly accessible cipher to be 'blessed' by a national agency such as the NSA. The release of its specification by NBS stimulated an explosion of public and academic interest in cryptography.

The aging DES was officially replaced by the Advanced Encryption Standard (AES) in 2001 when NIST announced FIPS 197. After an open competition, NIST selected Rijndael, submitted by two Belgian cryptographers, to be the AES. DES, and more secure variants of it (such as Triple DES), are still used today, having been incorporated into many national and organizational standards. However, its 56-bit key-size has been shown to be insufficient to guard against brute force attacks (one such attack, undertaken by the cyber civil-rights group Electronic Frontier Foundation in 1997, succeeded in 56 hours.[33]) As a result, use of straight DES encryption is now without doubt insecure for use in new cryptosystem designs, and messages protected by older cryptosystems using DES, and indeed all messages sent since 1976 using DES, are also at risk. Regardless of DES' inherent quality, the DES key size (56-bits) was thought to be too small by some even in 1976, perhaps most publicly by Whitfield Diffie. There was suspicion that government organizations even then had sufficient computing power to break DES messages; clearly others have achieved this capability.

Public key[edit]

The second development, in 1976, was perhaps even more important, for it fundamentally changed the way cryptosystems might work. This was the publication of the paper New Directions in Cryptography by Whitfield Diffie and Martin Hellman. It introduced a radically new method of distributing cryptographic keys, which went far toward solving one of the fundamental problems of cryptography, key distribution, and has become known as Diffie–Hellman key exchange. The article also stimulated the almost immediate public development of a new class of enciphering algorithms, the asymmetric key algorithms.

Prior to that time, all useful modern encryption algorithms had been symmetric key

algorithms, in which the same cryptographic key is used with the underlying algorithm by both the sender and the recipient, who must both keep it secret. All of the electromechanical machines used in World War II were of this logical class, as were the Caesar and Atbash ciphers and essentially all cipher systems throughout history. The 'key' for a code is, of course, the codebook, which must likewise be distributed and kept secret, and so shares most of the same problems in practice.
Of necessity, the key in every such system had to be exchanged between the communicating parties in some secure way prior to any use of the system (the term usually used is 'via a secure channel') such as a trustworthy courier with a briefcase handcuffed to a wrist, or face-to-face contact, or a loyal carrier pigeon. This requirement is never trivial and very rapidly becomes unmanageable as the number of participants increases, or when secure channels aren't available for key exchange, or when, as is sensible cryptographic practice, keys are frequently changed. In particular, if messages are meant to be secure from other users, a separate key is required for each possible pair of users. A system of this kind is known as a secret key, or symmetric key cryptosystem. D-H key exchange (and succeeding improvements and variants) made operation of these systems much easier, and more secure, than had ever been possible before in all of history.
In contrast, asymmetric key encryption uses a pair of mathematically related keys, each of which decrypts the encryption performed using the other. Some, but not all, of these algorithms have the additional property that one of the paired keys cannot be deduced from the other by any known method other than trial and error. An algorithm of this kind is known as a public key or asymmetric key system. Using such an algorithm, only one key pair is needed per user. By designating one key of the pair as private (always secret), and the other as public (often widely available), no secure channel is needed for key exchange. So long as the private key stays secret, the public key can be widely known for a very long time without compromising security, making it safe to reuse the same key pair indefinitely.
For two users of an asymmetric key algorithm to communicate securely over an insecure channel, each user will need to know their own public and private keys as well as the other user's public key. Take this basic scenario: Alice and Bob each have a pair of keys they've been using for years with many other users. At the start of their message, they exchange public keys, unencrypted over an insecure line. Alice then encrypts a message using her private key, and then re-encrypts that result using Bob's public key. The double-encrypted message is then sent as digital data over a wire from Alice to Bob. Bob receives the bit stream and decrypts it using his own private key, and then decrypts that bit stream using Alice's public key. If the final result is recognizable as a message, Bob can be confident that the message actually came from someone who knows Alice's private key (presumably actually her if she's been careful with her private key), and that anyone eavesdropping on the channel will need Bob's private key in order to understand the message.

Asymmetric algorithms rely for their effectiveness on a class of problems in mathematics called one-way functions, which require relatively little computational power to execute, but vast amounts of power to reverse, if reversal is possible at all. A classic example of a one-way function is multiplication of very large prime numbers. It's fairly quick to multiply two large primes, but very difficult to find the factors of the product of two large primes. Because of the mathematics of one-way functions, most possible keys are bad choices as cryptographic keys; only a small fraction of the possible keys of a given length are suitable, and so asymmetric algorithms require very long keys to reach the same level of security provided by relatively shorter symmetric keys. The need to both generate the key pairs, and perform the encryption/decryption operations make asymmetric algorithms computationally expensive, compared to most symmetric algorithms. Since symmetric algorithms can often use any sequence of (random, or at least unpredictable) bits as a key, a disposable *session key* can be quickly generated for short-term use. Consequently, it is common practice to use a long asymmetric key to exchange a disposable, much shorter (but just as strong) symmetric key. The slower asymmetric algorithm securely sends a symmetric session key, and the faster symmetric algorithm takes over for the remainder of the message.
Asymmetric key cryptography, Diffie–Hellman key exchange, and the best known of the public key / private key algorithms (i.e., what is usually called the RSA algorithm), all seem to have been independently developed at a UK intelligence agency before the public announcement by Diffie and Hellman in 1976. GCHQ has released documents claiming they had developed public key cryptography before the publication of Diffie and Hellman's paper.[*citation needed*] Various classified papers were written at GCHQ during the 1960s and 1970s which eventually led to schemes essentially identical to RSA encryption and to Diffie–Hellman key exchange in 1973 and 1974. Some of these have now been published, and the inventors (James H. Ellis, Clifford Cocks, and Malcolm Williamson) have made public (some of) their work.

Hashing[edit]

Hashing is a common technique used in cryptography to encode information quickly using typical algorithms. Generally, an algorithm is applied to a string of text, and the resulting string becomes the "hash value". This creates a "digital fingerprint" of the message, as the specific hash value is used to identify a specific message. The output from the algorithm is also referred to as a "message digest" or a "check sum". Hashing is good for determining if information has been changed in transmission. If the hash value is different upon reception than upon sending, there is evidence the message has been altered. Once the algorithm has been applied to the data to be hashed, the hash function produces a fixed-length output. Essentially, anything passed through the hash function should resolve to the same length output as anything else passed through the same hash function. It is important to note that hashing is not the same as encrypting. Hashing is a one-way operation that is used to

transform data into the compressed message digest. Additionally, the integrity of the message can be measured with hashing. Conversely, encryption is a two-way operation that is used to transform plaintext into cipher-text and then vice versa. In encryption, the confidentiality of a message is guaranteed.[34]
Hash functions can be used to verify digital signatures, so that when signing documents via the Internet, the signature is applied to one particular individual. Much like a hand-written signature, these signatures are verified by assigning their exact hash code to a person. Furthermore, hashing is applied to passwords for computer systems. Hashing for passwords began with the UNIX operating system. A user on the system would first create a password. That password would be hashed, using an algorithm or key, and then stored in a password file. This is still prominent today, as web applications that require passwords will often hash user's passwords and store them in a database.[35]

Cryptography politics[edit]

The public developments of the 1970s broke the near monopoly on high quality cryptography held by government organizations (see S Levy's *Crypto* for a journalistic account of some of the policy controversy of the time in the US). For the first time ever, those outside government organizations had access to cryptography not readily breakable by anyone (including governments). Considerable controversy, and conflict, both public and private, began more or less immediately, sometimes called the crypto wars. They have not yet subsided. In many countries, for example, export of cryptography is subject to restrictions. Until 1996 export from the U.S. of cryptography using keys longer than 40 bits (too small to be very secure against a knowledgeable attacker) was sharply limited. As recently as 2004, former FBI Director Louis Freeh, testifying before the 9/11 Commission, called for new laws against public use of encryption.
One of the most significant people favoring strong encryption for public use was Phil Zimmermann. He wrote and then in 1991 released PGP (Pretty Good Privacy), a very high quality crypto system. He distributed a freeware version of PGP when he felt threatened by legislation then under consideration by the US Government that would require backdoors to be included in all cryptographic products developed within the US. His system was released worldwide shortly after he released it in the US, and that began a long criminal investigation of him by the US Government Justice Department for the alleged violation of export restrictions. The Justice Department eventually dropped its case against Zimmermann, and the freeware distribution of PGP has continued around the world. PGP even eventually became an open Internet standard (RFC 2440 or OpenPGP).

Modern cryptanalysis[edit]

While modern ciphers like AES and the higher quality asymmetric ciphers are widely considered unbreakable, poor designs and implementations are still sometimes adopted and there have been important cryptanalytic breaks of deployed crypto

systems in recent years. Notable examples of broken crypto designs include the first Wi-Fi encryption scheme WEP, the Content Scrambling System used for encrypting and controlling DVD use, the A5/1 and A5/2 ciphers used in GSM cell phones, and the CRYPTO1 cipher used in the widely deployed MIFARE Classic smart cards from NXP Semiconductors, a spun off division of Philips Electronics. All of these are symmetric ciphers. Thus far, not one of the mathematical ideas underlying public key cryptography has been proven to be 'unbreakable', and so some future mathematical analysis advance might render systems relying on them insecure. While few informed observers foresee such a breakthrough, the key size recommended for security as best practice keeps increasing as increased computing power required for breaking codes becomes cheaper and more available. Quantum computers, if ever constructed with enough capacity, could break existing public key algorithms and efforts are underway to develop and standardize post-quantum cryptography.
Even without breaking encryption in the traditional sense, side-channel attacks can be mounted that exploit information gained from the way a computer system is implemented, such as cache memory usage, timing information, power consumption, electromagnetic leaks or even sounds emitted. Newer cryptographic algorithms are being developed that make such attacks more difficult.